UNIT–I: LEARNING AND MOTIVATION

1. LEARNING – CONCEPT AND FACTORS AFFECTING LEARNING

Meaning of Learning

Learning is a relatively permanent change in behaviour, knowledge, skills, attitudes, or understanding that occurs as a result of experience, practice, training, or interaction with the environment.

According to psychologists, learning is not merely acquiring information but also modifying behaviour and adapting to new situations.

Definitions of Learning

Hilgard:
"Learning is the process by which an activity originates or is changed through training procedures."

Crow and Crow:
"Learning is the acquisition of habits, knowledge, and attitudes."

Gates:
Learning is the modification of behaviour through experience and training.

Characteristics of Learning

1. Learning is a continuous process.
2. It results in a relatively permanent change in behaviour.
3. Learning involves experience and practice.
4. It is goal-directed.
5. Learning helps adaptation to the environment.
6. It may be positive or negative.
7. It involves cognitive, affective, and psychomotor domains.
8. Learning is universal and occurs throughout life.

Nature of Learning

- Learning begins at birth and continues throughout life.
- It helps in personal and social adjustment.
- Learning may occur intentionally or unintentionally.
- It involves interaction between the learner and environment.
- Learning improves efficiency and performance.

Factors Affecting Learning

Learning is influenced by several factors which can be classified into:

A. Learner-Related Factors

1. Intelligence

- Higher intelligence generally facilitates learning.
- Intelligent students understand concepts more quickly.

2. Motivation

- Motivation creates interest and willingness to learn.
- Highly motivated learners learn faster and retain more.

3. Attention

- Concentration and attention are essential for effective learning.
- Distractions reduce learning efficiency.

4. Interest

- Learners acquire knowledge better when interested in the subject.

5. Physical Health

- Good health promotes effective learning.
- Illness and fatigue reduce learning capacity.

6. Emotional Stability

- Positive emotions support learning.
- Anxiety, fear, and stress hinder learning.

7. Maturity

- Physical and mental maturity influence readiness to learn.

8. Previous Knowledge

- Prior experiences facilitate new learning through association.

B. Teacher-Related Factors

1. Teaching Method

- Appropriate methods improve understanding and retention.

2. Teacher's Personality

- Enthusiastic and supportive teachers enhance learning.

3. Subject Mastery

- A teacher's command over content affects instructional effectiveness.

4. Classroom Management

- Proper discipline and organization support learning.

5. Teacher-Student Relationship

- Positive relationships encourage participation and confidence.

C. Environmental Factors

1. Classroom Environment

- Proper lighting, ventilation, and seating facilitate learning.

2. Learning Resources

- Availability of books, laboratories, technology, and teaching aids improves learning.

3. Family Environment

- Supportive families encourage educational achievement.

4. Social Environment

- Peer groups and community influence learning behaviour.

D. Material-Related Factors

1. Nature of Subject Matter

- Simple and meaningful content is learned more easily.

2. Organization of Learning Material

- Well-structured content promotes understanding.

3. Difficulty Level

- Excessively difficult material discourages learners.

Educational Implications

1. Teachers should motivate learners.
2. Individual differences should be considered.
3. Learning environments should be conducive.
4. Teaching methods should be learner-centered.
5. Teaching aids should be effectively utilized.

2(a). APPROACHES TO LEARNING

I. COGNITIVE APPROACH – GESTALT THEORY

Meaning of Gestalt

The German word "Gestalt" means "whole", "configuration", or "organized pattern."

Gestalt psychologists believed that learning occurs through understanding relationships among different parts of a situation and perceiving them as a whole.

Major contributors:

- Max Wertheimer
- Kurt Koffka
- Wolfgang Kohler

Basic Principles of Gestalt Theory

1. Learning Through Insight

Insight means sudden understanding of relationships within a problem situation.

2. Whole is Greater Than the Sum of Parts

Learning occurs when the learner perceives the complete pattern rather than isolated elements.

3. Perception is Organized

Individuals naturally organize experiences into meaningful wholes.

4. Problem Solving Through Understanding

Learners solve problems by comprehending relationships rather than trial and error.

Kohler's Experiment

Experiment

- Kohler conducted experiments on chimpanzees.
- A banana was placed beyond reach.
- Sticks and boxes were available.
- After observing the situation, the chimpanzee suddenly combined sticks or stacked boxes and obtained the banana.

Conclusion

Learning occurred through insight rather than random attempts.

Characteristics of Insight Learning

1. Sudden understanding.
2. Based on perception of relationships.
3. Intelligent behaviour.
4. Transferable to similar situations.
5. Leads to meaningful learning.

Educational Implications of Gestalt Theory

1. Teaching should emphasize understanding.
2. Learning materials should be organized.
3. Problem-solving activities should be encouraged.
4. Students should discover relationships independently.
5. Whole-to-part teaching should be adopted.

II. BEHAVIOURIST APPROACH

Behaviourists explain learning as a change in observable behaviour resulting from stimulus-response associations.

Major contributors:

- Ivan Pavlov
- Edward Thorndike
- B.F. Skinner

A. Pavlov's Classical Conditioning Theory

Introduction

Ivan Pavlov, a Russian physiologist, discovered Classical Conditioning.

Experiment

Before Conditioning

- Food → Salivation
- Bell → No Salivation

During Conditioning

- Bell + Food → Salivation

After Conditioning

- Bell Alone → Salivation

Key Concepts

Unconditioned Stimulus (UCS)

Food

Unconditioned Response (UCR)

Salivation due to food

Conditioned Stimulus (CS)

Bell

Conditioned Response (CR)

Salivation to bell

Educational Implications

1. Positive classroom environment can promote learning.
2. Good habits can be developed through association.
3. Emotional responses can be conditioned.

B. Thorndike's Trial and Error Theory

Introduction

Edward L. Thorndike proposed that learning occurs through trial and error.

Puzzle Box Experiment

- A hungry cat was placed inside a puzzle box.
- Food was kept outside.
- Through repeated attempts, the cat learned how to escape.

Laws of Learning

1. Law of Readiness

Learning occurs best when learners are prepared.

2. Law of Exercise

Practice strengthens learning.

3. Law of Effect

Responses followed by satisfaction are strengthened.

Educational Implications

1. Practice and repetition are important.
2. Learning should be rewarding.
3. Student readiness must be considered.
4. Reinforcement strengthens learning.

C. Skinner's Operant Conditioning Theory

Introduction

B.F. Skinner developed Operant Conditioning.

Learning occurs through consequences of behaviour.

Skinner Box Experiment

- A rat was placed in a box.
- Pressing a lever provided food.
- Lever pressing increased because it was rewarded.

Key Concepts

Reinforcement

Any consequence that strengthens behaviour.

Positive Reinforcement

Giving a reward.

Example:
Praise, grades, certificates.

Negative Reinforcement

Removing unpleasant conditions.

Example:
Removing extra work after good performance.

Punishment

Consequence that decreases behaviour.

Positive Punishment

Adding unpleasant consequences.

Negative Punishment

Removing desirable privileges.

Educational Implications

1. Use rewards to encourage learning.
2. Reinforce desired behaviours immediately.
3. Provide feedback regularly.
4. Develop programmed learning materials.

III. SOCIAL COGNITIVE APPROACH – BANDURA

Introduction

Albert Bandura proposed Social Cognitive Theory.

Learning occurs through observation, imitation, and modelling.

Key Concepts

Observational Learning

Individuals learn by observing others.

Modelling

Learners imitate behaviours demonstrated by models.

Vicarious Reinforcement

Observing rewards received by others encourages imitation.

Self-Efficacy

Belief in one's ability to succeed.

Bandura's Bobo Doll Experiment

Procedure

- Children observed adults behaving aggressively toward a Bobo doll.
- Later children interacted with the doll.

Findings

- Children imitated the aggressive behaviour.

Conclusion

Behaviour can be learned through observation.

Processes of Observational Learning

1. Attention

Observing the model.

2. Retention

Remembering observed behaviour.

3. Reproduction

Performing the behaviour.

4. Motivation

Desire to imitate.

Educational Implications

1. Teachers serve as role models.
2. Demonstration method becomes effective.
3. Positive behaviours should be displayed.
4. Self-efficacy should be developed.
5. Cooperative learning should be encouraged.

2(b). RELEVANCE AND APPLICATIONS OF THE ABOVE APPROACHES TO LEARNING

Applications of Gestalt Theory

1. Promotes meaningful learning.
2. Encourages problem-solving abilities.
3. Supports discovery learning.
4. Develops creativity and critical thinking.
5. Enhances conceptual understanding.

Applications of Behaviourist Theory

1. Useful for habit formation.
2. Effective in classroom management.
3. Encourages skill acquisition.
4. Helps in behaviour modification.
5. Supports reinforcement-based teaching.

Applications of Social Cognitive Theory

1. Learning through demonstration.
2. Peer tutoring and collaborative learning.
3. Character education.
4. Development of self-confidence.
5. Use of role models and media learning.

Comparative Importance

Approach	Focus	Educational Value
Gestalt	Insight and understanding	Problem-solving and conceptual learning
Behaviourist	Observable behaviour	Skill development and behaviour modification
Social Cognitive	Observation and modelling	Social learning and self-efficacy

---

3. TRANSFER OF LEARNING

Meaning

Transfer of learning refers to the influence of previous learning on new learning or performance.

When knowledge, skills, habits, or experiences acquired in one situation affect learning in another situation, transfer of learning occurs.

Definitions

Crow and Crow:

Transfer of learning is the carry-over of habits, knowledge, skills, and attitudes from one learning situation to another.

Characteristics

1. Connects previous and new learning.
2. Makes learning more efficient.
3. Enhances problem-solving.
4. Improves educational effectiveness.

Types of Transfer of Learning

A. Positive Transfer

Previous learning facilitates new learning.

Example

- Learning Hindi helps learning Sanskrit.
- Knowledge of mathematics helps physics.

B. Negative Transfer

Previous learning interferes with new learning.

Example

- Driving on the left side may create difficulty when driving on the right side.

C. Zero Transfer

Previous learning has no effect on new learning.

Example

- Learning music may not affect learning chemistry.

D. Near Transfer

Transfer occurs between similar situations.

Example

- Learning addition helps learning subtraction.

E. Far Transfer

Transfer occurs between dissimilar situations.

Example

- Logical thinking learned in mathematics helps decision-making.

F. Vertical Transfer

Earlier learning helps mastery of advanced learning.

Example

- Arithmetic before algebra.

G. Horizontal Transfer

Transfer between subjects of equal complexity.

Example

- Knowledge of geography helping history.

Theories of Transfer

1. Theory of Formal Discipline

Mental faculties can be strengthened through practice.

2. Theory of Identical Elements

Transfer occurs when two situations share common elements.

3. Generalization Theory

Transfer depends on understanding general principles.

Teaching for Transfer

Meaning

Teaching for transfer means organizing instruction so that learners can apply knowledge and skills in new situations.

Methods of Teaching for Transfer

1. Provide Meaningful Learning

Understanding promotes transfer.

2. Emphasize General Principles

Teach rules and concepts.

3. Use Varied Situations

Expose students to different contexts.

4. Encourage Problem Solving

Develop transferable thinking skills.

5. Connect New and Previous Knowledge

Build associations.

6. Use Practical Applications

Relate lessons to real life.

7. Promote Reflective Thinking

Encourage analysis and evaluation.

Educational Significance

1. Saves time and effort.
2. Increases efficiency.
3. Encourages application of knowledge.
4. Supports lifelong learning.
5. Enhances creativity and adaptability.

4. MOTIVATION – CONCEPT AND SIGNIFICANCE

Meaning of Motivation

The word motivation is derived from the Latin word "Movere" which means "to move."

Motivation is the internal or external force that energizes, directs, and sustains behaviour toward achieving goals.

Definitions

Good:
Motivation is the process of arousing, sustaining, and regulating activity.

Woodworth:
Motivation is a state that initiates and directs behaviour.

Characteristics of Motivation

1. Goal-oriented.
2. Continuous process.
3. Influences behaviour.
4. Creates energy and enthusiasm.
5. Directs learning activities.

Significance of Motivation in Education

1. Increases Interest

Motivated students participate actively.

2. Improves Learning

Motivation enhances understanding and retention.

3. Increases Achievement

Students perform better academically.

4. Develops Self-Confidence

Motivated learners believe in their abilities.

5. Promotes Creativity

Encourages exploration and innovation.

6. Enhances Classroom Participation

Students become active learners.

7. Supports Persistence

Learners continue despite difficulties.

TYPES OF MOTIVATION

A. Intrinsic Motivation

Meaning

Motivation that arises from within the individual.

Learning is pursued because it is interesting, enjoyable, or satisfying.

Examples

- Reading for pleasure.
- Solving puzzles for enjoyment.
- Learning out of curiosity.

Characteristics

1. Self-directed.
2. Long-lasting.
3. Produces deeper learning.
4. Encourages creativity.

B. Extrinsic Motivation

Meaning

Motivation that comes from external rewards or pressures.

Examples

- Grades.
- Certificates.
- Praise.
- Rewards.

Characteristics

1. Reward-oriented.
2. Externally controlled.
3. Useful for short-term goals.
4. Encourages immediate action.

Difference Between Intrinsic and Extrinsic Motivation

Intrinsic Motivation	Extrinsic Motivation
Comes from within	Comes from outside
Based on interest	Based on rewards
Long-lasting	Often temporary
Encourages deep learning	Encourages performance

MASLOW'S HIERARCHY OF NEEDS

Introduction

Abraham Maslow proposed that human behaviour is motivated by a hierarchy of needs arranged from lower to higher levels.

Individuals attempt to satisfy lower needs before progressing to higher needs.

Hierarchy of Needs

1. Physiological Needs

Basic survival needs.

Examples:

- Food
- Water
- Sleep
- Shelter

2. Safety Needs

Need for security and protection.

Examples:

- Physical safety
- Health
- Stable environment

3. Love and Belongingness Needs

Need for affection and social relationships.

Examples:

- Friendship
- Family
- Group acceptance

4. Esteem Needs

Need for recognition and respect.

Examples:

- Achievement
- Status
- Self-respect

---

5. Self-Actualization Needs

Need to realize one's full potential.

Examples:

- Creativity
- Personal growth
- Self-fulfillment

Educational Implications of Maslow's Theory

1. Basic needs must be satisfied for effective learning.
2. Safe classrooms promote learning.
3. Positive teacher-student relationships are essential.
4. Recognition improves motivation.
5. Opportunities for creativity should be provided.

MOTIVATIONAL DEVICES FOR CLASSROOM TEACHING

Meaning

Motivational devices are techniques used by teachers to stimulate students' interest and participation in learning.

Important Motivational Devices

1. Praise and Appreciation

- Recognizes student efforts.
- Enhances confidence.

---

2. Rewards and Incentives

- Certificates.
- Badges.
- Prizes.

---

3. Interesting Teaching Methods

- Projects.
- Experiments.
- Games.

---

4. Use of Teaching Aids

- Charts.
- Models.
- Multimedia presentations.

---

5. Goal Setting

- Clear learning objectives motivate students.

---

6. Healthy Competition

- Encourages achievement and participation.

---

7. Questioning Technique

- Stimulates curiosity and thinking.

---

8. Activity-Based Learning

- Promotes active involvement.

---

9. Feedback and Reinforcement

- Helps students improve performance.

---

10. Relating Content to Real Life

- Makes learning meaningful and relevant.

---

11. Cooperative Learning

- Encourages teamwork and peer support.

---

12. Providing Success Experiences

- Small achievements build confidence and motivation.

UNIT – II : INDIVIDUAL DIFFERENCES AND COGNITIVE PROCESSES

1. INDIVIDUAL DIFFERENCES – NATURE, TYPES, CAUSES AND ACCOMMODATING INDIVIDUAL DIFFERENCES IN CLASSROOM

Meaning of Individual Differences

Individual differences refer to the variations or distinctions among individuals in physical, mental, emotional, social, intellectual, and personality characteristics.

No two individuals are exactly alike. Even twins differ in abilities, interests, attitudes, and behaviour. These differences influence the learning process and educational achievement.

Definitions

Skinner:
Individual differences refer to variations among individuals in various traits and characteristics.

James Drever:
Individual differences are the differences which distinguish one person from another.

Nature of Individual Differences

The important characteristics of individual differences are:

1. Universal Phenomenon

Individual differences exist among all people regardless of age, gender, culture, or nationality.

2. Quantitative and Qualitative

Differences may be measured numerically (intelligence, height, weight) or observed qualitatively (attitudes, creativity).

3. Dynamic in Nature

Differences may change due to experience, education, and environment.

4. Hereditary and Environmental

Individual differences result from both genetic inheritance and environmental influences.

5. Present in All Areas

Differences occur in physical, intellectual, emotional, social, and moral aspects.

6. Educationally Significant

These differences affect learning speed, achievement, interests, and classroom behaviour.

Types of Individual Differences

Individual differences can be categorized as follows:

A. Physical Differences

Differences in:

- Height
- Weight
- Body structure
- Health
- Physical strength

Educational Implications

Teachers should consider physical limitations and provide suitable activities.

B. Intellectual Differences

Differences in:

- Intelligence
- Reasoning ability
- Problem-solving skills
- Learning capacity

Educational Implications

Different instructional strategies should be used for learners with varying intellectual abilities.

C. Emotional Differences

Differences in:

- Emotional maturity
- Emotional stability
- Anxiety levels
- Self-control

Educational Implications

Emotionally balanced classrooms enhance learning.

D. Social Differences

Differences in:

- Social skills
- Leadership qualities
- Cooperation
- Communication

Educational Implications

Group activities help develop social skills.

E. Personality Differences

Differences in:

- Introversion
- Extroversion
- Self-confidence
- Behaviour patterns

Educational Implications

Teachers should respect different personality traits.

F. Interest Differences

Students vary in:

- Academic interests
- Hobbies
- Vocational preferences

Educational Implications

Instruction should connect with student interests.

G. Aptitude Differences

Differences in special abilities such as:

- Musical aptitude
- Artistic aptitude
- Mechanical aptitude
- Mathematical aptitude

Educational Implications

Students should be guided according to their aptitudes.

H. Learning Differences

Differences occur in:

- Learning speed
- Learning styles
- Retention ability

Educational Implications

Teaching methods should be flexible.

Causes of Individual Differences

Individual differences arise due to several factors:

A. Heredity

Heredity refers to traits transmitted from parents to children through genes.

Examples

- Intelligence
- Physical features
- Temperament

Educational Importance

Teachers should recognize inherited abilities and limitations.

B. Environment

Environment includes all external influences affecting development.

Examples

- Family
- School
- Society
- Peer group

Educational Importance

A supportive environment enhances learning.

C. Family Background

Differences arise because of:

- Economic status
- Parental education
- Home atmosphere

D. Educational Opportunities

Differences in access to:

- Schools
- Books
- Technology
- Learning resources

E. Cultural Factors

Culture influences:

- Values
- Beliefs
- Language
- Behaviour

F. Health and Nutrition

Healthy children generally perform better academically.

G. Motivation and Interest

Motivated learners learn more effectively than unmotivated learners.

Accommodating Individual Differences in Classroom

Meaning

Accommodating individual differences means adapting teaching methods, learning experiences, and classroom environments to meet diverse learner needs.

Strategies for Accommodating Individual Differences

1. Individualized Instruction

Teaching should be adapted according to student abilities.

2. Differentiated Teaching

Different tasks and activities should be provided for different learners.

3. Flexible Grouping

Students can be grouped according to:

- Ability

- Interest
- Learning style

4. Remedial Teaching

Special assistance should be provided to slow learners.

5. Enrichment Programmes

Advanced learners should receive challenging activities.

6. Activity-Based Learning

Hands-on experiences accommodate diverse learning styles.

7. Continuous Assessment

Regular assessment helps identify learner needs.

8. Guidance and Counselling

Students should receive support for academic and personal issues.

9. Use of Educational Technology

Multimedia and digital tools support diverse learners.

10. Positive Classroom Climate

Respect and acceptance should be encouraged.

Educational Significance

1. Promotes inclusive education.
2. Improves learning outcomes.
3. Enhances student participation.
4. Reduces learning difficulties.
5. Supports holistic development.

2. UNDERSTANDING DIFFERENCES BASED ON COGNITIVE ABILITIES IN CHILDREN WITH LEARNING DIFFICULTIES

Meaning of Learning Difficulties

Learning difficulties refer to conditions that interfere with a student's ability to acquire, process, understand, or use information effectively.

Children with learning difficulties usually have average or above-average intelligence but face problems in specific academic areas.

Characteristics of Children with Learning Difficulties

Academic Characteristics

- Difficulty in reading
- Difficulty in writing
- Difficulty in mathematics
- Poor memory
- Slow learning speed

Behavioural Characteristics

- Lack of confidence
- Frustration
- Low motivation
- Poor concentration

Social Characteristics

- Withdrawal from peers
- Difficulty in communication
- Low participation

A. Slow Learner

Meaning

A slow learner is a child who learns at a slower pace than average students and requires more time and assistance.

Generally, slow learners possess below-average intellectual ability but are not intellectually disabled.

Characteristics of Slow Learners

Academic Characteristics

- Slow understanding
- Difficulty grasping abstract concepts
- Poor academic performance
- Limited vocabulary

Behavioural Characteristics

- Low confidence
- Lack of initiative
- Dependency on others

Cognitive Characteristics

- Slow information processing
- Weak memory
- Poor problem-solving ability

Causes of Slow Learning

Biological Factors

- Poor health
- Neurological problems

Environmental Factors

- Poverty
- Lack of educational support

Psychological Factors

- Low motivation
- Emotional disturbances

Educational Strategies for Slow Learners

1. Individual Attention

Provide personal guidance.

2. Simplified Instruction

Use simple language and examples.

3. Repetition and Practice

Repeated exercises strengthen learning.

4. Use of Teaching Aids

Visual and concrete materials improve understanding.

5. Positive Reinforcement

Encourage small achievements.

6. Remedial Teaching

Provide additional support sessions.

B. Dyslexia

Meaning

Dyslexia is a specific learning disability that primarily affects reading, spelling, and language processing abilities.

It is neurological in origin and is unrelated to intelligence.

Definition

According to the International Dyslexia Association:

Dyslexia is a specific learning disability characterized by difficulties with accurate and fluent word recognition, spelling, and decoding abilities.

Characteristics of Dyslexia

Reading Difficulties

- Slow reading
- Incorrect pronunciation
- Difficulty recognizing words

Writing Difficulties

- Poor spelling
- Letter reversals
- Writing errors

Language Difficulties

- Difficulty understanding language sounds
- Problems with vocabulary

Cognitive Difficulties

- Weak phonological awareness
- Difficulty processing language information

Common Signs of Dyslexia

- Confusion between similar letters
- Reading slowly
- Omitting words while reading
- Difficulty following written instructions

Educational Strategies for Dyslexic Learners

1. Multisensory Teaching

Use visual, auditory, and kinesthetic methods.

2. Phonics-Based Instruction

Teach sound-symbol relationships systematically.

3. Assistive Technology

Use audiobooks and text-to-speech software.

4. Extra Time

Allow additional time during tests.

5. Positive Support

Avoid criticism and encourage effort.

6. Individualized Education Plans

Provide customized learning support.

Comparison Between Slow Learner and Dyslexic Learner

Basis	Slow Learner	Dyslexic Learner
Intelligence	Below average	Average or above average
Learning Problem	General learning difficulty	Specific reading difficulty
Academic Performance	Weak in most subjects	Mainly affected in reading and writing
Learning Speed	Slow overall	Normal except language-related tasks
Support Required	General academic support	Specialized reading intervention

Educational Implications

1. Early identification is essential.
2. Inclusive classrooms should be promoted.
3. Individualized teaching strategies are necessary.
4. Positive reinforcement improves confidence.
5. Parents and teachers should work together.

3(a). COGNITIVE PROCESS – CONCEPT AND PROCESSES

Meaning of Cognitive Process

Cognitive processes are mental activities involved in acquiring, organizing, storing, retrieving, and using information.

These processes help individuals understand and interact with their environment.

Major Cognitive Processes

1. Sensation
2. Perception
3. Attention
4. Memory
5. Concept Formation
6. Problem Solving

1. SENSATION

Meaning

Sensation is the process through which sensory organs receive stimuli from the environment and transmit them to the brain.

It is the first step in acquiring knowledge.

Characteristics

- Basic cognitive process
- Involves sensory receptors
- Provides raw information

Types of Sensations

Visual Sensation

Obtained through eyes.

Auditory Sensation

Obtained through ears.

Olfactory Sensation

Obtained through nose.

Gustatory Sensation

Obtained through tongue.

Tactile Sensation

Obtained through skin.

Importance in Learning

Learning begins with sensory experiences.

2. PERCEPTION

Meaning

Perception is the process of organizing, interpreting, and giving meaning to sensory information.

Characteristics

- Meaningful interpretation of stimuli
- Depends on previous experience
- Selective in nature

Factors Affecting Perception

Internal Factors

- Interest
- Motivation
- Experience

External Factors

- Intensity
- Size
- Contrast
- Novelty

Importance in Learning

Perception helps learners understand their environment.

3. ATTENTION

Meaning

Attention is the concentration of mental activity on a particular object, idea, or task.

Characteristics

- Selective
- Limited
- Goal-oriented

Types of Attention

Voluntary Attention

Controlled by the individual.

Involuntary Attention

Automatically attracted by stimuli.

Habitual Attention

Developed through practice.

Factors Affecting Attention

Objective Factors

- Size
- Colour
- Movement
- Intensity

Subjective Factors

- Interest
- Motivation
- Needs

Importance in Learning

Attention is essential for effective learning and retention.

4. MEMORY

Meaning

Memory is the mental process of retaining and recalling past experiences and information.

Processes of Memory

1. Encoding

Receiving information.

2. Storage

Retaining information.

3. Retrieval

Recalling information.

Types of Memory

Sensory Memory

Stores information briefly.

Short-Term Memory

Temporary storage.

Long-Term Memory

Permanent storage.

Importance in Learning

Memory enables retention and application of knowledge.

---

5. CONCEPT FORMATION

Meaning

Concept formation is the process of grouping objects, events, or ideas based on common characteristics.

Examples

- Animal
- Plant
- Democracy
- Triangle

Stages of Concept Formation

1. Observation

Observing objects and events.

2. Comparison

Identifying similarities and differences.

3. Classification

Grouping items.

4. Generalization

Developing concepts.

Importance in Learning

Concept formation promotes understanding and classification of knowledge.

---

6. PROBLEM SOLVING

Meaning

Problem solving is the process of overcoming obstacles and finding solutions to achieve goals.

Characteristics

- Goal-directed
- Cognitive activity
- Requires reasoning

Steps in Problem Solving

1. Identifying the Problem

Recognizing the difficulty.

2. Understanding the Problem

Gathering relevant information.

3. Generating Alternatives

Developing possible solutions.

4. Evaluating Alternatives

Selecting the best option.

5. Implementing the Solution

Applying the chosen solution.

6. Evaluating Results

Checking effectiveness.

Importance in Learning

Problem solving develops critical thinking and decision-making skills.

3(b). EDUCATIONAL IMPLICATIONS OF COGNITIVE PROCESSES

Educational Implications of Sensation

1. Use multisensory teaching methods.
2. Provide concrete experiences.
3. Use visual and auditory aids.
4. Ensure healthy sensory functioning.

Educational Implications of Perception

1. Relate new learning to prior experiences.
2. Use meaningful examples.
3. Organize learning materials clearly.
4. Eliminate misleading stimuli.

Educational Implications of Attention

1. Use attractive teaching aids.

2. Create interest in lessons.
3. Avoid monotony.
4. Use questioning techniques.
5. Maintain classroom discipline.

Educational Implications of Memory

1. Encourage repetition and practice.
2. Use mnemonic devices.
3. Organize information logically.
4. Promote meaningful learning.
5. Conduct regular revision.

Educational Implications of Concept Formation

1. Use concrete examples before abstract ideas.
2. Encourage observation and classification.
3. Promote discovery learning.
4. Use inductive teaching methods.

Educational Implications of Problem Solving

1. Encourage inquiry and exploration.
2. Present real-life problems.
3. Develop critical thinking skills.
4. Promote independent learning.
5. Use project-based learning.

UNIT – III : TEACHING AND TEACHING PROCESS

PART A : TEACHING

1. CONCEPT, NATURE AND PRINCIPLES OF TEACHING

Concept of Teaching

Teaching is a systematic process of facilitating learning by guiding, motivating, directing, and helping learners acquire knowledge, skills, attitudes, values, and desirable behaviour.

Teaching is not merely the transmission of information but a dynamic interaction between teacher and learners aimed at bringing about desirable changes in behaviour.

Definitions of Teaching

H.C. Morrison

Teaching is an intimate contact between a more mature personality and a less mature one designed to further the education of the latter.

John Dewey

Teaching is the process of arranging and managing situations in which students actively learn.

Smith

Teaching is a system of actions intended to produce learning.

Nature of Teaching

Teaching possesses several important characteristics:

1. Teaching is a Social Process

Teaching occurs through interaction between teacher and students.

2. Teaching is Goal-Oriented

Every teaching activity is directed towards achieving predetermined objectives.

3. Teaching is Interactive

It involves communication and cooperation between teacher and learners.

4. Teaching is Both Art and Science

- Art because it requires creativity and skill.
- Science because it follows principles and systematic procedures.

5. Teaching is Dynamic

Teaching changes according to learner needs, curriculum, and societal demands.

6. Teaching is Developmental

It promotes intellectual, emotional, social, and moral development.

7. Teaching is Continuous

The teaching-learning process continues throughout life.

8. Teaching is Planned

Effective teaching requires planning and organization.

9. Teaching is Diagnostic and Remedial

Teachers identify learning difficulties and provide solutions.

10. Teaching is Learner-Centered

Modern teaching focuses on learner participation and activity.

Principles of Teaching

The following principles guide effective teaching:

1. Principle of Motivation

Learning becomes effective when learners are motivated.

2. Principle of Readiness

Teaching should consider the learner's readiness and maturity.

3. Principle of Activity

Learning occurs best through active participation.

4. Principle of Individual Differences

Teaching should accommodate diverse learner needs.

5. Principle of Interest

Interesting learning experiences increase participation.

6. Principle of Reinforcement

Positive reinforcement strengthens learning.

7. Principle of Correlation

Knowledge should be linked with other subjects and experiences.

8. Principle of Feedback

Feedback helps learners improve performance.

9. Principle of Learning by Doing

Practical experiences improve understanding.

10. Principle of Democratic Participation

Students should be encouraged to express opinions freely.

Educational Significance

1. Enhances learning effectiveness.
2. Promotes student participation.
3. Improves classroom interaction.
4. Facilitates achievement of educational objectives.
5. Supports holistic development.

---

2. RELATIONSHIP BETWEEN TEACHING AND LEARNING

Meaning of Learning

Learning refers to a relatively permanent change in behaviour, knowledge, skills, attitudes, or understanding resulting from experience and practice.

Relationship Between Teaching and Learning

Teaching and learning are complementary processes.

Teaching facilitates learning, while learning is the outcome of effective teaching.

Without learning, teaching has not achieved its purpose.

Characteristics of the Relationship

1. Interdependent

Teaching and learning depend upon each other.

2. Goal-Oriented

Both aim at achieving educational objectives.

3. Interactive

Continuous interaction occurs between teacher and learner.

4. Dynamic

Both processes change according to circumstances and learner needs.

5. Continuous

Teaching and learning continue throughout life.

Comparison Between Teaching and Learning

Basis	Teaching	Learning
Meaning	Process of facilitating learning	Process of acquiring knowledge and skills
Performer	Teacher	Learner
Nature	External guidance	Internal change
Purpose	Promote learning	Gain knowledge and development
Outcome	Instruction	Behavioural change

Importance of Relationship

1. Teaching promotes learning.
2. Learning indicates teaching effectiveness.
3. Both contribute to educational success.
4. They support student development.
5. Effective teaching leads to meaningful learning.

3. LEVELS, PHASES AND COMPONENTS OF TEACHING

Levels of Teaching

Teaching occurs at different levels according to learner participation and intellectual development.

Morris L. Bigge proposed three levels:

1. Memory Level
2. Understanding Level
3. Reflective Level

A. Memory Level Teaching

Meaning

Teaching focused on memorization and recall of facts.

Characteristics

- Teacher-centered
- Emphasis on rote learning
- Low learner participation
- Focus on factual knowledge

Advantages

- Useful for basic information.
- Helps acquisition of foundational knowledge.

Limitations

- Encourages rote memorization.
- Does not develop higher thinking.

B. Understanding Level Teaching

Meaning

Teaching aimed at understanding concepts, principles, and relationships.

Characteristics

- Emphasis on comprehension.
- Learner participation increases.
- Development of reasoning.

Advantages

- Promotes meaningful learning.
- Enhances understanding.

Limitations

- Requires skilled teachers.
- More time-consuming.

C. Reflective Level Teaching

Meaning

Highest level of teaching emphasizing critical thinking and problem-solving.

Characteristics

- Student-centered

- Encourages inquiry
- Develops creativity
- Promotes independent thinking

Advantages

- Enhances critical thinking.
- Develops problem-solving ability.

Limitations

- Requires mature learners.
- Difficult to implement in large classes.

Comparison of Levels of Teaching

Aspect	Memory Level	Understanding Level	Reflective Level
Focus	Memorization	Understanding	Reflection
Teacher Role	Dominant	Guide	Facilitator
Learner Role	Passive	Active	Highly Active
Thinking Level	Low	Moderate	High

Phases of Teaching

Teaching consists of three phases:

1. Pre-active Phase
2. Interactive Phase
3. Post-active Phase

A. Pre-active Phase

Meaning

Planning stage before classroom teaching.

Activities

- Setting objectives
- Selecting content
- Choosing methods
- Preparing teaching aids

B. Interactive Phase

Meaning

Actual classroom teaching phase.

Activities

- Presentation
- Questioning
- Discussion
- Classroom interaction

C. Post-active Phase

Meaning

Evaluation phase after teaching.

Activities

- Assessment
- Feedback
- Remediation
- Reflection

Components of Teaching

Teaching involves four major components:

1. Teacher

Functions

- Planner
- Guide
- Facilitator
- Evaluator

Importance

Teacher directs and organizes learning experiences.

---

2. Student

Functions

- Active learner
- Participant
- Problem solver

Importance

Student is the central focus of teaching.

---

3. Teaching-Learning Material (TLM)

Examples

- Textbooks
- Charts
- Models
- Audio-visual aids
- Computers

Importance

Makes learning meaningful and effective.

---

4. Classroom Climate

Meaning

The emotional and social atmosphere of the classroom.

Characteristics

- Safety
- Cooperation
- Respect
- Supportiveness

Importance

Positive classroom climate promotes learning.

---

4. INTERRELATEDNESS OF OBJECTIVES, TEACHING-LEARNING EXPERIENCES AND EVALUATION

Meaning of Educational Objectives

Objectives are desired learning outcomes expected from students.

Meaning of Teaching-Learning Experiences

Activities and experiences provided to achieve objectives.

Examples:

- Lectures
- Discussions
- Experiments
- Projects

Meaning of Evaluation

Process of determining the extent to which objectives have been achieved.

Interrelationship

The three components are closely related.

Objectives

↓

Determine

↓

Teaching-Learning Experiences

↓

Produce

↓

Learning Outcomes

↓

Measured By

↓

Evaluation

Explanation

Step 1: Objectives

Specify what students should learn.

Step 2: Learning Experiences

Provide opportunities for achieving objectives.

Step 3: Evaluation

Measures whether objectives have been achieved.

Example

Objective

Students will understand photosynthesis.

Learning Experience

Demonstration and discussion.

Evaluation

Written test and practical activity.

Educational Importance

1. Ensures systematic teaching.
2. Improves instructional effectiveness.
3. Facilitates assessment.
4. Enhances learning outcomes.
5. Promotes accountability.

---

5. CONTENT ANALYSIS AND TASK ANALYSIS

A. Content Analysis

Meaning

Content analysis is the process of examining subject matter and organizing it into meaningful units for teaching.

Objectives of Content Analysis

1. Organize content logically.
2. Identify key concepts.
3. Determine learning objectives.
4. Facilitate lesson planning.

Steps in Content Analysis

1. Selection of Content

Choose relevant subject matter.

2. Identification of Concepts

Identify important ideas.

3. Sequencing

Arrange content logically.

4. Determination of Objectives

Specify learning outcomes.

5. Evaluation Planning

Design assessment procedures.

Importance

1. Improves lesson organization.
2. Enhances understanding.

3. Facilitates effective teaching.
4. Supports curriculum planning.

B. Task Analysis

Meaning

Task analysis is the process of breaking a complex task into smaller and manageable steps.

Objectives

1. Identify task requirements.
2. Organize learning activities.
3. Facilitate skill acquisition.

Steps in Task Analysis

1. Identify the Task

Define the learning goal.

2. Break Into Subtasks

Divide task into smaller components.

3. Sequence Steps

Arrange steps logically.

4. Determine Performance Criteria

Specify expected performance.

Importance

1. Simplifies complex learning.
2. Helps skill development.
3. Supports individualized instruction.
4. Facilitates assessment.

PART B : TEACHING PROCESS

1. TEACHING TECHNOLOGY: CONCEPT, ASSUMPTIONS, CHARACTERISTICS AND COMPONENTS

Concept of Teaching Technology

Teaching Technology refers to the systematic application of scientific knowledge, psychological principles, methods, techniques, and resources to improve teaching and learning.

It focuses on making teaching more effective, efficient, and objective-oriented.

Definitions

G.O.M. Leith

Teaching technology is the application of scientific knowledge to teaching.

B.C. Rai

Teaching technology is a systematic way of designing, implementing, and evaluating the teaching-learning process.

Assumptions of Teaching Technology

1. Teaching is a Scientific Process

Teaching can be planned and improved systematically.

2. Learning Outcomes Can Be Measured

Student achievement can be assessed objectively.

3. Behaviour Can Be Modified

Teaching can bring desirable behavioural changes.

4. Objectives Guide Instruction

Teaching should focus on achieving objectives.

5. Technology Improves Effectiveness

Use of modern techniques enhances learning.

Characteristics of Teaching Technology

1. Goal-oriented.
2. Scientific and systematic.
3. Student-centered.
4. Objective-based.
5. Uses feedback.
6. Employs modern technology.
7. Facilitates evaluation.

Components of Teaching Technology

A. Planning

Activities

- Formulation of objectives
- Content selection
- Method selection
- Preparation of materials

B. Organization

Activities

- Classroom management
- Instructional implementation
- Resource utilization

C. Evaluation

Activities

- Assessment
- Feedback
- Improvement of instruction

Importance of Teaching Technology

1. Improves teaching quality.
2. Enhances learning outcomes.
3. Saves time and effort.

4. Promotes active learning.
5. Facilitates evaluation.

2. CRITERIA OF EFFECTIVE TEACHING AND METHODS OF ASSESSMENT OF TEACHING

Criteria of Effective Teaching

Effective teaching is teaching that successfully promotes learning and student development.

Characteristics of Effective Teaching

1. Clear Objectives

Teaching should have specific goals.

2. Subject Mastery

Teacher should possess content knowledge.

3. Effective Communication

Concepts should be presented clearly.

4. Student Participation

Learners should be actively involved.

5. Appropriate Methods

Methods should suit learner needs.

6. Classroom Management

Maintain discipline and order.

7. Use of Teaching Aids

Teaching aids improve understanding.

8. Continuous Assessment

Monitor student progress regularly.

9. Feedback

Provide constructive feedback.

10. Positive Classroom Climate

Promote cooperation and respect.

Methods of Assessment of Teaching

A. Classroom Observation

Meaning

Systematic observation of teacher performance during instruction.

Advantages

1. Direct assessment.
2. Immediate feedback.
3. Objective evaluation.

Limitations

1. Observer bias.
2. Teacher anxiety.

B. Peer Assessment

Meaning

Evaluation of teaching by fellow teachers.

Advantages

1. Professional feedback.
2. Sharing of best practices.

Limitations

1. Personal bias.
2. Lack of objectivity.

C. Self-Reporting

Meaning

Teacher evaluates own teaching performance.

Methods

- Self-reflection
- Diaries
- Checklists

Advantages

1. Encourages professional growth.
2. Promotes reflective practice.

Limitations

1. Subjective.
2. May lack accuracy.

D. Evaluation by a Supervisor

Meaning

Assessment conducted by principal, headmaster, or educational supervisor.

Advantages

1. Expert evaluation.
2. Accountability.

Limitations

1. Anxiety among teachers.
2. Occasional bias.

3. TEACHER BEHAVIOUR DURING TEACHING: FLANDERS' INTERACTION ANALYSIS SYSTEM (FIAS)

Introduction

Ned A. Flanders developed the Flanders Interaction Analysis System (FIAS) in 1960.

It is a technique used to observe and analyze classroom verbal interaction between teachers and students.

Meaning

FIAS studies classroom communication by categorizing teacher talk, student talk, and silence.

Objectives of FIAS

1. Analyze classroom interaction.
2. Improve teaching effectiveness.
3. Identify teacher behaviour patterns.
4. Enhance student participation.

Categories of FIAS

Flanders classified classroom behaviour into ten categories.

Teacher Talk (Categories 1–7)

Teacher talk usually occupies the largest portion of classroom communication.

Indirect Teacher Influence

Category 1: Accepts Feelings

Teacher accepts student emotions.

Category 2: Praises or Encourages

Teacher motivates students.

Category 3: Accepts or Uses Ideas of Students

Teacher values student contributions.

Category 4: Asks Questions

Teacher stimulates thinking through questioning.

Direct Teacher Influence

Category 5: Lecturing

Teacher provides information.

Category 6: Giving Directions

Teacher gives instructions.

Category 7: Criticizing or Justifying Authority

Teacher controls behaviour or defends authority.

Student Talk (Categories 8–9)

Category 8: Student Talk Response

Students respond to teacher questions.

Category 9: Student Talk Initiation

Students initiate ideas and discussions.

Silence or Confusion

Category 10

Periods of silence, confusion, or pauses.

Diagrammatic Representation

Teacher Talk

1. Accepts Feelings
2. Praises
3. Accepts Ideas
4. Questions
5. Lectures
6. Directions
7. Criticizes

↓

Student Talk

1. Response
2. Initiation

↓

1. Silence/Confusion

Advantages of FIAS

1. Provides objective classroom analysis.
2. Improves teacher effectiveness.
3. Encourages reflective teaching.
4. Promotes student participation.
5. Useful in teacher training.

Limitations of FIAS

1. Focuses mainly on verbal behaviour.
2. Ignores non-verbal communication.
3. Time-consuming.
4. Requires trained observers.

Educational Importance

1. Helps improve classroom interaction.
2. Encourages democratic teaching.
3. Identifies strengths and weaknesses.
4. Improves teaching strategies.
5. Enhances student engagement.

UNIT – IV : MODELS AND APPROACHES OF TEACHING

1. CONCEPT OF MODELS OF TEACHING

Meaning of Teaching Model

A model of teaching is a systematic pattern or plan that guides teachers in organizing teaching-learning activities to achieve specific educational objectives.

Teaching models provide a framework for planning curriculum, designing instructional materials, selecting teaching strategies, and evaluating learning outcomes.

They describe how teaching should be conducted in order to bring about desired behavioural changes in learners.

Definitions of Models of Teaching

Bruce Joyce and Marsha Weil

A model of teaching is a pattern or plan that can be used to shape a curriculum, design instructional materials, and guide classroom instruction.

B.K. Passi

A teaching model is a generalized pattern of teaching behaviour designed to achieve specific educational objectives.

Characteristics of Models of Teaching

1. Goal-Oriented

Every model aims to achieve specific educational objectives.

2. Systematic

Models provide a structured sequence of teaching activities.

3. Scientific

They are based on psychological and educational theories.

4. Flexible

Models can be modified according to learner needs.

5. Student-Centered

Most modern models emphasize active learner participation.

6. Practical

They help teachers organize effective instruction.

7. Replicable

The same model can be used repeatedly in similar situations.

Importance of Models of Teaching

1. Improve Teaching Effectiveness

Models help teachers teach systematically.

2. Facilitate Learning

Students understand concepts more effectively.

3. Promote Active Participation

Many models encourage learner involvement.

4. Develop Higher Mental Abilities

Models enhance reasoning, creativity, and problem-solving.

5. Guide Curriculum Planning

Teachers can organize content efficiently.

Educational Significance

1. Makes teaching scientific.
2. Provides instructional guidance.
3. Improves classroom interaction.
4. Enhances learning outcomes.
5. Supports professional development of teachers.

---

2. ELEMENTS OF MODELS OF TEACHING

According to Joyce and Weil, every teaching model contains four major elements.

---

1. Syntax

Meaning

Syntax refers to the sequence or stages of teaching activities within a model.

It explains how teaching proceeds step by step.

Example

In an inquiry model:

- Present problem
- Gather information
- Form hypothesis
- Test hypothesis
- Draw conclusion

Importance

Provides systematic organization of teaching.

---

2. Social System

Meaning

The social system describes the roles of teacher and students and the nature of classroom interaction.

Components

- Teacher role
- Student role

- Classroom atmosphere
- Communication pattern

Importance

Creates a supportive learning environment.

3. Principles of Reaction

Meaning

These are the ways teachers respond to student behaviour and learning activities.

Examples

- Providing feedback
- Encouraging participation
- Reinforcing correct responses

Importance

Guides teacher behaviour during instruction.

4. Support System

Meaning

The support system includes resources and materials required for implementing the model.

Examples

- Textbooks
- Audio-visual aids
- Computers
- Laboratories
- Worksheets

Importance

Provides necessary resources for effective teaching.

Summary of Elements

Element	Meaning
Syntax	Sequence of teaching activities
Social System	Roles and interactions
Principles of Reaction	Teacher responses
Support System	Instructional resources

3. FAMILIES OF MODELS OF TEACHING

Meaning

Teaching models are grouped into different families based on their objectives and theoretical foundations.

Bruce Joyce and Marsha Weil classified teaching models into four major families.

1. Information Processing Family

Meaning

Focuses on intellectual development and cognitive processes.

Objectives

- Develop thinking skills
- Enhance concept formation
- Improve problem-solving

Examples

- Inquiry Training Model
- Concept Attainment Model
- Advance Organizer Model

Characteristics

- Learner-centered
- Cognitive development oriented
- Encourages inquiry

2. Personal Family

Meaning

Focuses on personal growth, self-development, and emotional development.

Objectives

- Self-awareness
- Creativity
- Self-esteem

Examples

- Non-directive Teaching Model
- Synectics Model

Characteristics

- Individual development
- Self-directed learning
- Emotional growth

3. Social Family

Meaning

Emphasizes social interaction and cooperative learning.

Objectives

- Social skills
- Democratic participation
- Teamwork

Examples

- Group Investigation Model
- Role Playing Model

Characteristics

- Cooperative learning
- Group interaction
- Social competence

4. Behavioural Systems Family

Meaning

Based on behaviour modification principles.

Objectives

- Behaviour change
- Skill development
- Habit formation

Examples

- Direct Instruction Model
- Programmed Instruction Model

Characteristics

- Reinforcement-oriented
- Observable behaviour
- Structured learning

Comparison of Families

Family	Main Focus	Examples
Information Processing	Cognitive development	Inquiry Training, Concept Attainment
Personal	Personal growth	Synectics, Non-directive Teaching
Social	Social interaction	Role Playing, Group Investigation
Behavioural Systems	Behaviour modification	Direct Instruction

4. TYPES OF MODELS OF TEACHING

A. RICHARD SUCHMAN'S INQUIRY TRAINING MODEL

Introduction

Richard Suchman developed the Inquiry Training Model based on scientific inquiry.

The model trains students to investigate problems systematically and discover solutions independently.

Objectives

1. Develop inquiry skills.
2. Promote scientific thinking.
3. Enhance problem-solving ability.
4. Encourage curiosity.

Assumptions

1. Natural Curiosity

Humans naturally seek explanations.

2. Inquiry Can Be Taught

Scientific thinking can be developed through practice.

3. Learning Through Discovery

Knowledge is acquired through investigation.

Syntax (Steps)

Phase 1: Presenting the Problem

Teacher presents a puzzling situation.

Phase 2: Data Gathering and Verification

Students collect relevant information.

Phase 3: Experimentation

Students test hypotheses.

Phase 4: Formulating Explanations

Students explain findings.

Phase 5: Analysis of Inquiry Process

Students reflect on their investigation.

Advantages

1. Develops critical thinking.
2. Promotes independent learning.
3. Encourages scientific attitude.
4. Increases curiosity.

Limitations

1. Time-consuming.
2. Requires skilled teachers.
3. Difficult in large classes.

Educational Applications

- Science teaching
- Research projects
- Problem-solving activities

B. GLASER'S BASIC TEACHING MODEL

Introduction

Robert Glaser proposed the Basic Teaching Model as a systematic framework for instruction.

The model focuses on planning, teaching, and evaluation.

Components of Glaser's Model

1. Instructional Objectives

Specify intended learning outcomes.

2. Entering Behaviour

Assess learners' prior knowledge and readiness.

3. Instructional Procedures

Select methods and activities for teaching.

4. Performance Assessment

Evaluate learner achievement.

Diagrammatic Representation

Instructional Objectives

↓

Entering Behaviour

↓

Instructional Procedures

↓

Performance Assessment

Advantages

1. Systematic planning.
2. Objective-based teaching.
3. Continuous evaluation.
4. Improves learning effectiveness.

Educational Importance

1. Helps lesson planning.
2. Supports individualized instruction.
3. Facilitates assessment.

C. INFORMATION PROCESSING MODEL

Meaning

The Information Processing Model explains learning as the processing of information through various mental stages.

It is based on cognitive psychology.

Basic Assumption

The human mind functions like a computer by receiving, processing, storing, and retrieving information.

Components

1. Input

Information enters through sensory organs.

2. Processing

Information is interpreted and organized.

3. Storage

Information is stored in memory.

4. Retrieval

Stored information is recalled when needed.

Stages of Information Processing

Sensory Memory

Receives sensory information.

↓

Short-Term Memory

Temporary storage and processing.

↓

Long-Term Memory

Permanent storage.

↓

Retrieval

Recall of information.

Characteristics

1. Cognitive approach.
2. Focus on thinking processes.
3. Emphasizes memory and understanding.
4. Promotes meaningful learning.

Advantages

1. Improves comprehension.
2. Enhances memory.
3. Develops cognitive skills.
4. Supports meaningful learning.

Educational Applications

1. Use of advance organizers.
2. Concept mapping.
3. Memory strategies.
4. Problem-solving activities.

D. CONCEPT ATTAINMENT MODEL

Introduction

The Concept Attainment Model was developed by Jerome Bruner.

The model helps learners identify concepts through examples and non-examples.

Meaning

Concept attainment is the process of discovering common attributes of a concept by comparing examples.

Objectives

1. Develop analytical thinking.
2. Improve concept formation.
3. Enhance classification skills.
4. Promote inductive reasoning.

Assumptions

1. Learners Can Discover Concepts

Students are capable of identifying patterns.

2. Concept Learning Improves Thinking

Concept formation enhances reasoning ability.

Syntax (Steps)

Phase 1: Presentation of Data

Teacher presents examples and non-examples.

Phase 2: Testing Attainment

Students identify characteristics.

Phase 3: Analysis of Thinking Strategy

Students explain their reasoning.

Example

Concept: Triangle

Examples

- Equilateral triangle
- Isosceles triangle
- Scalene triangle

Non-examples

- Circle
- Rectangle
- Square

Students identify common characteristics and formulate the concept.

Advantages

1. Promotes concept formation.
2. Encourages active participation.
3. Develops reasoning.
4. Improves classification skills.

Limitations

1. Requires careful planning.
2. Time-consuming.
3. Difficult for younger learners.

Educational Applications

1. Mathematics
2. Science
3. Language learning
4. Social sciences

5. APPROACHES TO TEACHING

Meaning of Teaching Approach

A teaching approach is a broad philosophy or method that guides the teaching-learning process.

It determines how teaching is organized and how learners interact with knowledge.

A. PARTICIPATORY APPROACH

Meaning

Participatory Approach emphasizes active involvement of learners in teaching-learning activities.

Students become partners in the learning process.

Characteristics

1. Learner participation.
2. Democratic environment.
3. Collaboration.
4. Shared decision-making.
5. Active learning.

Methods Used

- Discussion
- Seminar
- Debate
- Group work
- Project work

Advantages

1. Increases motivation.
2. Improves communication skills.
3. Develops leadership qualities.
4. Enhances learning retention.

Use in Teaching

1. Group projects.
2. Cooperative learning.
3. Classroom discussions.
4. Peer learning activities.

B. CHILD-CENTERED APPROACH

Meaning

Child-centered approach places the child at the center of the teaching-learning process.

Teaching is organized according to learner needs, interests, and abilities.

Characteristics

1. Learner-centered.
2. Activity-oriented.
3. Flexible curriculum.
4. Individual differences considered.
5. Learning by doing.

Principles

1. Respect for individuality.
2. Freedom of expression.
3. Self-learning.
4. Active participation.

Advantages

1. Promotes creativity.
2. Enhances confidence.
3. Encourages independent learning.
4. Develops problem-solving skills.

Use in Teaching

1. Activity-based learning.
2. Project method.
3. Discovery learning.
4. Experiential learning.

C. CONSTRUCTIVIST APPROACH

Meaning

Constructivism views learning as an active process in which learners construct knowledge through experience and interaction.

Major contributors:

- Jean Piaget
- Lev Vygotsky
- Jerome Bruner

Characteristics

1. Learner constructs knowledge.
2. Active participation.
3. Prior knowledge is important.
4. Collaborative learning.
5. Problem-solving orientation.

Principles

Learning is Active

Students actively engage in learning.

Knowledge is Constructed

Learners create understanding through experience.

Social Interaction Supports Learning

Interaction facilitates knowledge construction.

Advantages

1. Meaningful learning.
2. Critical thinking development.
3. Problem-solving enhancement.
4. Greater learner autonomy.

Use in Teaching

1. Inquiry-based learning.
2. Project work.
3. Group discussions.
4. Real-life problem solving.

D. INVESTIGATORY APPROACH

Meaning

Investigatory Approach emphasizes inquiry, exploration, and systematic investigation.

Students discover knowledge through observation, experimentation, and research.

Characteristics

1. Inquiry-oriented.
2. Problem-centered.
3. Research-based.
4. Active participation.
5. Scientific thinking.

Steps

1. Identification of Problem

Recognize a problem or question.

2. Data Collection

Gather information.

3. Hypothesis Formation

Develop possible explanations.

4. Investigation

Test hypotheses.

5. Conclusion

Draw conclusions.

Advantages

1. Develops scientific attitude.
2. Promotes critical thinking.
3. Encourages curiosity.
4. Improves research skills.

Use in Teaching

1. Science experiments.
2. Social science projects.
3. Field studies.
4. Research assignments.

COMPARISON OF TEACHING APPROACHES

Approach	Focus	Role of Teacher	Role of Student
Participatory	Participation	Facilitator	Active participant
Child-Centered	Learner needs	Guide	Central figure
Constructivist	Knowledge construction	Facilitator	Knowledge constructor
Investigatory	Inquiry and research	Mentor	Investigator

UNIT – V : TEACHING AS A PROFESSION

1. DEFINITION AND CHARACTERISTICS OF A PROFESSION

Meaning of Profession

A profession is an occupation that requires specialized knowledge, advanced education, specific skills, training, and adherence to ethical standards for providing services to society.

A profession is more than a job because it involves commitment, responsibility, expertise, and service to humanity.

Definitions of Profession

Abraham Flexner

A profession is an occupation involving intellectual operations with large individual responsibility.

Cogan

A profession is a vocation whose practice is founded upon an understanding of the theoretical structure of some department of learning or science.

Characteristics of a Profession

1. Specialized Knowledge

A profession requires a well-defined body of knowledge.

Example

Doctors study medicine, lawyers study law, and teachers study education.

2. Specialized Training

Professionals undergo systematic education and training.

Example

Teachers complete teacher education programmes such as B.Ed., D.El.Ed., M.Ed., etc.

3. Service Orientation

A profession primarily aims at serving society.

Example

Teachers contribute to the intellectual and moral development of students.

4. Professional Competence

Professionals possess specific skills and expertise.

Example

Teachers require instructional, communication, and evaluation skills.

5. Code of Ethics

Every profession follows ethical principles and standards.

Example

Teachers maintain honesty, fairness, and impartiality.

6. Professional Organization

Professionals are guided by professional bodies and organizations.

Example

NCTE, NCERT, UGC, and teacher associations.

7. Social Recognition

Society recognizes professionals for their expertise and contribution.

8. Autonomy

Professionals enjoy a degree of independence in performing their duties.

9. Continuous Professional Development

Professionals continuously update their knowledge and skills.

10. Accountability

Professionals are responsible for the quality of their work.

Importance of Profession

1. Provides specialized services.
2. Promotes social welfare.
3. Ensures quality performance.
4. Establishes ethical standards.
5. Encourages professional growth.

2. CHARACTERISTICS OF TEACHING PROFESSION

Meaning of Teaching Profession

Teaching profession refers to the occupation of educating, guiding, and developing learners through specialized knowledge, professional skills, and ethical practices.

Teaching is considered a noble profession because teachers shape the future of individuals and society.

Characteristics of Teaching Profession

1. Service-Oriented Profession

Teaching aims at the welfare and development of learners.

Importance

Teachers contribute to nation-building and social progress.

2. Specialized Knowledge

Teachers require knowledge of:

- Subject matter
- Educational psychology
- Teaching methods
- Evaluation techniques

3. Professional Training

Teachers undergo formal preparation through teacher education programmes.

Examples

- B.Ed.
- D.El.Ed.
- M.Ed.

4. Continuous Learning

Teachers update knowledge through:

- Workshops
- Seminars
- Research
- Training programmes

5. Ethical Responsibility

Teachers are expected to follow professional ethics.

6. Intellectual Activity

Teaching involves thinking, planning, reasoning, and decision-making.

7. Social Responsibility

Teachers influence students, families, and communities.

8. Democratic Nature

Teaching promotes equality, freedom, and respect.

9. Human Relations

Teaching requires effective interaction with:

- Students
- Parents
- Colleagues
- Community members

10. Lifelong Commitment

Teaching requires dedication and continuous professional improvement.

Why Teaching is Called a Noble Profession

1. Shapes Future Citizens

Teachers prepare responsible citizens.

2. Promotes Social Change

Teachers help eliminate ignorance and social evils.

3. Builds Character

Teachers develop values and ethics.

4. Supports National Development

Education contributes to economic and social progress.

Educational Significance

1. Develops human resources.
2. Promotes cultural transmission.
3. Encourages social harmony.
4. Supports democratic values.
5. Contributes to national development.

3. PROFESSIONAL ETHICS FOR TEACHERS

Meaning of Professional Ethics

Professional ethics refer to the moral principles, values, standards, and rules that guide the behaviour and conduct of teachers.

Professional ethics help teachers perform their duties responsibly and maintain public trust.

Objectives of Professional Ethics

1. Promote professionalism.
2. Ensure quality education.
3. Protect student welfare.
4. Maintain dignity of the profession.
5. Encourage responsible conduct.

Professional Ethics for Teachers

A. Ethics Towards Students

1. Respect Individual Differences

Teachers should treat all students fairly.

2. Maintain Equality

Avoid discrimination based on:

- Caste
- Religion
- Gender

- Language
- Economic status

3. Ensure Student Welfare

Protect students physically, emotionally, and academically.

4. Encourage Learning

Create a supportive learning environment.

5. Maintain Confidentiality

Protect student information and records.

B. Ethics Towards Profession

1. Maintain Professional Competence

Continuously improve knowledge and skills.

2. Uphold Professional Dignity

Behave in a manner that enhances the reputation of teaching.

3. Follow Educational Policies

Respect institutional rules and regulations.

4. Demonstrate Commitment

Perform duties sincerely and responsibly.

C. Ethics Towards Colleagues

1. Maintain Cooperation

Work collaboratively with colleagues.

2. Show Respect

Respect professional opinions and contributions.

3. Avoid Unprofessional Behaviour

Avoid gossip, jealousy, and unfair criticism.

D. Ethics Towards Parents

1. Maintain Communication

Inform parents about student progress.

2. Build Positive Relationships

Work together for student development.

E. Ethics Towards Society

1. Promote Social Values

Encourage justice, equality, and democracy.

2. Serve Community Interests

Participate in community development activities.

3. Promote National Integration

Develop unity and patriotism among students.

Importance of Professional Ethics

1. Enhances public trust.
2. Improves educational quality.
3. Promotes professionalism.
4. Protects student interests.
5. Maintains dignity of teaching.

4. STRENGTHENING TEACHING PROFESSION

Meaning

Strengthening the teaching profession means improving the quality, status, effectiveness, and professional competence of teachers through training, support, policies, and professional development opportunities.

A. ROLE OF EDUCATIONAL ORGANIZATIONS IN THE PROFESSIONAL DEVELOPMENT OF TEACHERS

1. Role of UGC (University Grants Commission)

Meaning

The UGC is a statutory body responsible for coordination, determination, and maintenance of standards in higher education in India.

Entity

University Grants Commission

Functions Related to Teacher Development

1. Promotes Higher Education

Supports quality improvement in universities.

2. Provides Financial Assistance

Funds research and faculty development programmes.

3. Organizes Orientation Programmes

Enhances teaching competence.

4. Supports Research

Encourages teachers to undertake research projects.

5. Promotes Academic Excellence

Maintains standards in higher education.

2. Role of NCTE (National Council for Teacher Education)

Entity

National Council for Teacher Education

Functions

1. Regulates Teacher Education

Sets standards for teacher education institutions.

2. Develops Teacher Education Curriculum

Ensures quality teacher preparation.

3. Maintains Professional Standards

Improves teacher competence.

4. Accreditation and Recognition

Recognizes teacher education programmes.

5. Encourages Innovation

Promotes modern teaching practices.

---

3. Role of NCERT (National Council of Educational Research and Training)

Entity

National Council of Educational Research and Training

Functions

1. Curriculum Development

Prepares school curriculum frameworks.

2. Teacher Training

Conducts workshops and training programmes.

3. Research Activities

Undertakes educational research.

4. Development of Teaching Materials

Produces textbooks and instructional resources.

5. Educational Innovation

Introduces innovative teaching approaches.

---

4. Role of Universities

Entity

Universities

Functions

1. Teacher Preparation

Offer teacher education programmes.

2. Research Opportunities

Encourage educational research.

3. Professional Development

Conduct seminars and conferences.

4. Academic Leadership

Promote educational excellence.

---

5. Role of SIERT

Meaning

State Institute of Educational Research and Training.

Entity

State Institute of Educational Research and Training

Functions

1. Teacher Training

Provides in-service training.

2. Educational Research

Conducts state-level educational studies.

3. Curriculum Support

Assists curriculum implementation.

4. Development of Educational Materials

Produces instructional resources.

5. Innovation in Education

Promotes modern teaching practices.

B. ROLE OF TEACHER EDUCATION INSTITUTIONS IN THE PROFESSIONAL DEVELOPMENT OF TEACHERS

Meaning

Teacher Education Institutions prepare and develop teachers through pre-service and in-service education programmes.

Major Roles

1. Professional Preparation

Provide theoretical and practical knowledge.

---

2. Skill Development

Develop teaching, communication, and evaluation skills.

---

3. Teaching Practice

Provide classroom experience through internships.

---

4. Research Orientation

Encourage educational research.

---

5. Professional Values

Develop ethical and professional attitudes.

---

6. Continuing Education

Offer refresher courses and training programmes.

---

7. Technological Competence

Train teachers in educational technology.

Educational Importance

1. Improves teacher quality.
2. Promotes innovation.
3. Enhances classroom effectiveness.
4. Encourages professional growth.

C. ROLE OF SCHOOL AND COMMUNITY IN ENRICHING TEACHING PROFESSION

Role of School

1. Supportive Environment

Schools provide opportunities for professional growth.

2. Professional Development Activities

Organize workshops and seminars.

3. Collaborative Learning

Promote teamwork among teachers.

4. Performance Feedback

Help teachers improve instructional practices.

5. Innovation and Experimentation

Encourage use of new methods and technologies.

Role of Community

1. Community Participation

Supports educational activities.

2. Resource Support

Provides materials and expertise.

3. Recognition of Teachers

Improves professional status.

4. Cooperation with Schools

Enhances educational effectiveness.

5. Promoting Educational Awareness

Encourages public support for education.

Educational Significance

1. Strengthens teacher effectiveness.
2. Promotes community-school partnership.
3. Enhances educational quality.
4. Improves professional status of teachers.

5. BALANCING PERSONAL ASPIRATIONS AND PROFESSIONAL OBLIGATIONS BY TEACHERS

Meaning

Teachers, like all individuals, have personal goals, ambitions, family responsibilities, and interests. At the same time, they have professional obligations towards students, schools, and society.

Balancing personal aspirations and professional obligations means maintaining harmony between personal life and professional responsibilities without neglecting either.

Personal Aspirations of Teachers

Examples

- Higher education
- Career advancement
- Research activities
- Financial security
- Family responsibilities
- Personal well-being
- Social recognition

Professional Obligations of Teachers

Responsibilities Towards Students

- Quality teaching
- Guidance
- Evaluation
- Moral development

Responsibilities Towards School

- Maintaining discipline
- Following policies
- Participating in institutional activities

Responsibilities Towards Society

- Promoting values
- Supporting national development
- Encouraging responsible citizenship

Challenges in Balancing Personal and Professional Life

1. Workload

Excessive responsibilities may create stress.

2. Time Management Issues

Difficulty allocating time for family and profession.

3. Professional Expectations

High expectations from schools and society.

4. Career Pressures

Need for promotions and professional growth.

5. Emotional Stress

Managing students and personal responsibilities simultaneously.

Strategies for Balancing Personal Aspirations and Professional Obligations

1. Effective Time Management

Plan and prioritize tasks.

---

2. Continuous Professional Development

Enhance professional competence without neglecting personal goals.

---

3. Setting Realistic Goals

Maintain achievable personal and professional objectives.

---

4. Maintaining Work-Life Balance

Allocate adequate time for:

- Family
- Health
- Recreation
- Professional responsibilities

---

5. Stress Management

Practice healthy coping strategies.

---

6. Professional Commitment

Remain dedicated to student welfare and educational objectives.

---

7. Lifelong Learning

Continue self-improvement while fulfilling professional duties.

Educational Significance

1. Improves teacher effectiveness.
2. Enhances job satisfaction.
3. Reduces stress and burnout.
4. Promotes professional growth.
5. Improves student outcomes.
